SCI-NOI v0.1

SCI-NOI v0.1 r0.2 Scientific Rationale

Implementation-blind Stage B draft

SCI-NOI_SCIENTIFIC_RATIONALE v0.1/draft-r0.2

Not owner-accepted or frozen. No implementation conformity, validation, calibration, physical-noise, covariance-completeness, significance, performance, readiness, or production claim.

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Status: implementation-blind Stage B draft; not owner-accepted or frozen.

Normative authority: the six ordered modules in SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.2, binding-file SHA-256 5c6f954b457a546abcf74a1dc6dae190f2b22ea43c14edf34d2e4d2a8a704268. This rationale explains that authority and does not independently author an equation, requirement, assumption, definition, or prediction.

1. Scientific purpose

SCI-NOI separates three questions that are often accidentally merged. GEN defines a finite randomized realization ensemble from an immutable scientific parent. UNC defines an empirical quantity for one exact admitted ensemble. STD joins one immutable real-observation signal to one compatible positive conditional scale. Keeping the operations separate prevents a realization from silently becoming uncertainty, a reciprocal from silently becoming a weight, or a standardized map from silently becoming significance.

2. Why the ordinary route is fixed

The selected ordinary route changes one thing: a detector/channel sign at the exact PTC-to-frozen-MAP boundary. Every other scientifically consequential quantity stays fixed. The canonical equations make this auditably literal: the sign appears once, only on the PTC signal occurrence; the unsigned MAP normalization and all projection, coverage, response, support, validity, and publication gates remain the parent operation. The all-+1 result remains an ordinary MAP product, while each signed result is an NOI realization.

This distinction answers a causal question. Variation across members is due to the declared assignment law conditional on one fixed parent/operator. It is not variation from repeated observations, relearned state, a changed map operator, or newly drawn physical detector noise.

3. What network balance can and cannot establish

The owner selected network-stratified coefficient-mass balance and a complement-symmetric target-law family. This supports zero marginal detector sign mean and, through the fixed linear operator, zero target-law member mean. It does not guarantee that any individual member is source-free, that source structure is absent from squares or tails, or that a finite sample has count balance or independent members.

The detailed finite design remains a distinct scientific decision. ODQ-102D authorizes an implementation-blind proposal, not advance acceptance of its mechanics. The candidate design is therefore documented as one indivisible owner question. Population edge cases, exact tolerance, base law, bounded search, replacement, duplicate/complement treatment, rank, weights, count domain, keying, and product scope remain unavailable until answered.

4. Why the initial UNC quantity is narrow

For an accepted complement-symmetric design, the conditional target mean is known to be zero. The initial pointwise quantity therefore uses the design-weighted second moment about that known center. It does not subtract the finite ensemble mean and does not invoke a B-1 correction. It is calculated only where every admitted member supplies a valid finite value.

The result is a conditional detector-sign-randomization marginal second moment/variance for one fixed parent and operator. It may retain source-shaped and structured-residual imprint. It is not repeated-observation physical-noise variance, total MAP uncertainty, a complete covariance model, precision, or a tail calibration. Counts, sign diversity, complement-orbit diversity, sign rank, projected rank, effective Monte Carlo information, and estimator uncertainty remain separate facts.

5. Why reciprocal and standardization roles stay separate

A reciprocal of the pointwise second moment is algebraically available only on its finite strictly positive domain, but algebra does not make it inverse variance, precision, validity, exposure, or a consumer weight. r0.2 therefore retires the misleading active phrase "inverse ... scale" and asks the owner to approve a narrowly named reciprocal successor plus immutable profile/source rebinding. Until then the successor is unavailable.

The initial standardized product is equally narrow: the immutable real MAP signal divided by the positive square root of the compatible conditional second moment. Its unit is one. Dimensionlessness alone supplies no Gaussian, Student, pivotal, z, N-sigma, probability, detection, false-alarm, completeness, purity, or catalog interpretation.

6. Persistence, transformations, and profiles

Persisted ensembles, deterministic regeneration, and streaming sufficient statistics preserve different future questions. The owner-approved plan must say exactly what can be reconstructed, regenerated, or inferred; a composite is another plan and no fallback is implicit.

Transforms, Wiener operations, and FRUIT operations remain externally owned. Fixed application requires exact external authority and parity. Learning, selection, continuation, update, or member-specific replay creates new generations and cannot mix with the fixed ensemble.

The four Stage A profile byte strings are immutable and registered under the exact r0.18 binding. r0.2 changes notation and proposed action spelling, so it does not infer that those existing bytes evaluate the revised contract. An immutable successor binding is required before affected r0.2 evaluation.

7. Claim ceiling

This draft defines reviewable scientific authority and evidence obligations. It reports no implementation conformity, numerical validation, calibration, physical-noise validity, covariance completeness, Gaussian significance, achieved performance, readiness, freeze, production suitability, or production authorization.